MILITARY SPECIFICATION SHEET

WAVEGUIDES, RIGID, RECTANGULAR, REDUCED HEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the waveguides described herein shall consist of this specification sheet and the latest issue of MIL-W-85.

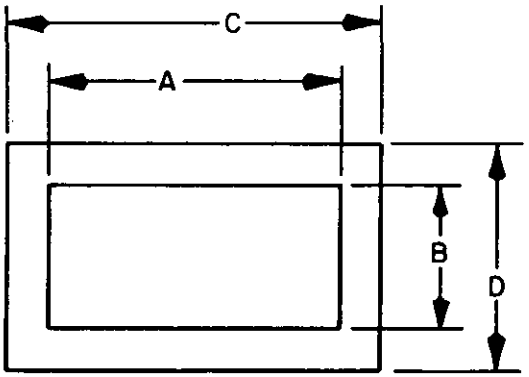


FIGURE 1. Waveguides, rectangular, reduced height.

REQUIREMENTS:

Material: See table I.

Design, construction, and physical dimensions: See figure 1 and table I.

Part number: See table I.

ENGINEERING INFORMATION:

Cross index and engineering information: See table II.

Typical resistivity and conductivity of materials: See table III.

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:44Z
Check the source to verify that this is the current version before use.

TABLE I. Waveguides, rigid, rectangular.

Part 2/ 3/ number M85/4-	Dimensions (inches) 1/											ASTM type material
	Inside			Outside			Wall		Max inner corner radius	Outer corner radius		
	Width (A)	Height (B)	Tolerance	Width (C)	Height (D)	Tolerance	Thickness (Nominal)	Max allow eccent 4/		Min	Max	
001-XXX 002-XXX	2.840 (72.14)	1.004 (25.50)	±.004 (.10)	3.000 (76.20)	1.164 (29.57)	±.004 (.10)	.080 (2.03)	.005 (.13)	.050 (1.27)	.040 (1.02)	.055 (1.40)	Copper alloy Aluminum alloy 6061
003-XXX 004-XXX	1.372 (34.85)	.487 (12.37)	±.003 (.08)	1.500 (38.10)	.615 (15.62)	±.003 (.08)	.064 (1.63)	.004 (.10)	.030 (.76)	.030 (.76)	.040 (1.02)	Copper alloy Aluminum alloy 6061
007-XXX	2.840 (72.14)	1.004 (25.50)	±.004 (.10)	3.000 (76.20)	1.164 (29.57)	±.004 (.10)	.080 (2.03)	.005 (.13)	.050 (1.27)	.040 (1.02)	.055 (1.40)	Aluminum alloy 6063
008-XXX	1.372 (34.85)	.487 (12.37)	±.003 (.08)	1.500 (38.10)	.615 (15.62)	±.003 (.08)	.064 (1.63)	.004 (.10)	.030 (.76)	.030 (.76)	.040 (1.02)	Aluminum alloy 6063
009-XXX 010-XXX 011-XXX	9.750 (247.65)	2.436 (61.87)	±.010 (.25)	10.000 (254.00)	2.686 (68.22)	±.010 (.25)	.125 (3.18)	.010 (.25)	.030 (.76)	.060 (1.52)	.080 (2.03)	Copper alloy Aluminum alloy 6061 Aluminum alloy 6063
012-XXX 013-XXX 014-XXX	9.750 (247.65)	2.436 (61.87)	±.010 (.25)	10.250 (260.35)	2.936 (74.57)	±.010 (.25)	.250 (6.35)	.010 (.25)	.030 (.76)	.060 (1.52)	.080 (2.03)	Copper alloy Aluminum alloy 6061 Aluminum alloy 6063
015-XXX 016-XXX 017-XXX	2.840 (72.14)	.670 (17.02)	±.005 (.13)	3.000 (76.20)	.830 (21.08)	±.005 (.13)	.080 (2.03)	.005 (.13)	.030 (.76)	.040 (1.02)	.055 (1.40)	Copper alloy Aluminum alloy 6061 Aluminum alloy 6063
018-XXX 019-XXX 020-XXX	1.872 (47.55)	.372 (9.45)	±.003 (.08)	2.000 (50.80)	.500 (12.70)	±.005 (.13)	.064 (1.63)	.004 (.10)	.030 (.76)	.030 (.76)	.050 (1.27)	Copper OF Aluminum alloy 6061 Aluminum alloy 6063
021-XXX 022-XXX	9.750 (247.65)	2.436 (61.87)	±.010 (.25)	10.125 (257.18)	2.811 (71.40)	±.010 (.25)	.1875 (4.760)	.010 (.25)	.030 (.76)	.060 (1.52)	.080 (2.03)	Aluminum alloy 6061 Aluminum alloy 6063
023-XXX 024-XXX	1.668 (42.37)	.506 (12.85)	±.005 (.13)	1.796 (45.62)	.634 (16.10)	±.005 (.13)	.064 (1.63)	.005 (.13)	.030 (.76)	.030 (.76)	.035 (.89)	Aluminum alloy 6061 Aluminum alloy 6063
025-XXX 026-XXX	1.020 (25.91)	.255 (6.48)	±.003 (.08)	1.148 (29.16)	.383 (29.16)	±.005 (.13)	.064 (1.63)	.003 (.08)	.016 (.41)	.016 (.41)	.031 (.79)	Aluminum alloy 6061 Aluminum alloy 6063
027-XXX 028-XXX 029-XXX	.965 (24.51)	.320 (8.13)	±.005 (.13)	1.083 (27.51)	.438 (11.13)	±.005 (.13)	.059 (1.50)	.005 (.13)	.030 (.76)	.030 (.76)	.035 (.89)	Copper alloy Aluminum alloy 6061 Aluminum alloy 6063
030-XXX 031-XXX	1.122 (28.50)	.249 (6.32)	±.004 (.10)	1.250 (31.75)	.376 (9.55)	±.004 (.10)	.064 (1.63)	.004 (.10)	.031 (.79)	.016 (.41)	.031 (.79)	Aluminum alloy 6061 Aluminum alloy 6063

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:44Z
Check the source to verify that this is the current version before use.

TABLE I. Waveguides, rigid, rectangular - Continued.

Part 2/ 3/ number M85/4-	Dimensions (inches) 1/											ASTM type material
	Inside			Outside			Wall		Max inner corner radius	Outer corner radius		
	Width (A)	Height (B)	Tolerance	Width (C)	Height (D)	Tolerance	Thickness (Nominal)	Max allow eccent 4/		Min	Max	
032-XXX	.847 (21.51)	.312 (7.92)	±.005 (.13)	.975 (24.76)	.440 (11.18)	±.005 (.13)	.064 (1.63)	.005 (.13)	.030 (.76)	.011 (.28)	.021 (.53)	Aluminum alloy 6061
033-XXX												Aluminum alloy 6063
034-XXX	.900 (22.86)	.200 (5.08)	±.004 (.10)	1.000 (25.40)	.300 (7.62)	±.004 (.10)	.050 (1.27)	.004 (.10)	.031 (.79)	.016 (.41)	.031 (.79)	Aluminum alloy 6061
035-XXX												Aluminum alloy 6063
036-XXX	.900 (22.86)	.200 (5.08)	±.004 (.10)	.960 (24.38)	.260 (6.60)	±.004 (.10)	.030 (.76)	.004 (.10)	.031 (.79)	.016 (.41)	.031 (.79)	Aluminum alloy 6061
037-XXX												Aluminum alloy 6063

1/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm. Metric equivalents are in parentheses.
2/ The complete part number shall be the number shown and a three-digit number indicating length in inches (for example, M85/4-001-096).
3/ Part numbers M85/4-005 and -006 have been transferred to MIL-W-85/1. For future acquisition use part numbers M85/1-155 and -158.
4/ The eccentricity of a waveguide is defined as one-half the difference between opposite wall thickness as measured at any cross section perpendicular to the longitudinal axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:44Z
Check the source to verify that this is the current version before use.

TABLE II. Cross index and engineering information.

New part number M85/4-	Old part number M85/1-	Old type designation	Recommended frequency range GHz TE_{10} mode	Cut-off for TE_{10} mode (GHz)	Theoretical 1/ attenuation lowest to highest frequency (dB/100 ft)	Theoretical 2/ peak power rating, lowest to highest frequency (megawatts)	Theoretical 3/ cw power rating lowest to highest frequency (kilowatts)
001-XXX	153-XXX	RG-109/U	2.60- 5.85 4/	2.080	1.131-0.947	5.720- 8.911	10.253- 12.245
002-XXX	---	---	2.60- 5.85 4/	2.080	1.145-0.959	5.720- 8.911	10.124- 12.091
003-XXX	154-XXX	RG-110/U	5.85-12.40 5/	4.285	3.399-2.819	1.577- 2.174	1.912- 2.305
004-XXX	---	---	5.85-12.40 5/	4.285	3.442-2.854	1.577- 2.174	1.888- 2.276
007-XXX	---	---	2.60- 5.85 4/	2.080	1.056-0.884	5.720- 8.911	10.98 - 13.11
008-XXX	---	---	5.85-12.40 5/	4.285	3.174-2.665	1.577- 2.174	2.048- 2.469
009-XXX	---	---	.75- 1.12	.605	.181- .134	46.886-66.763	158.612-214-245
010-XXX	---	---	.75- 1.12	.605	.272- .202	46.886-66.763	106.367-143.226
011-XXX	---	---	.75- 1.12	.605	.251- .186	46.886-66.763	115.266-155.547
012-XXX	---	---	.75- 1.12	.605	.181- .134	46.886-66.763	164.020-223.269
013-XXX	---	---	.75- 1.12	.605	.272- .202	46.886-66.763	109.993-148.109
014-XXX	---	---	.75- 1.12	.605	.251- .186	46.886-66.673	119.196-160.850
015-XXX	---	---	2.60- 3.95	2.080	1.187- .897	3.879- 5.497	9.203- 12.273
016-XXX	---	---	2.60- 3.95	2.080	1.784-1.348	3.879- 5.497	6.171- 8.167
017-XXX	---	---	2.60- 3.95	2.080	1.645-1.243	3.879- 5.497	6.692- 8.856
018-XXX	---	---	3.95- 5.85	3.155	2.530-1.958	1.427- 2.039	3.059- 3.953
019-XXX	---	---	3.95- 5.85	3.155	3.803-2.943	1.427- 2.039	2.051- 2.651
020-XXX	---	---	3.95- 5.85	3.155	3.506-2.274	1.427- 2.039	2.225- 3.431
021-XXX	---	---	.75- 1.12	.605	.272- .202	46.886-66.763	108.098-145.762
022-XXX	---	---	.75- 1.12	.605	.251- .186	46.886-66.763	108.098-145.762
023-XXX	---	---	4.3 -10.5	3.540	2.907-2.416	1.602- 2.657	2.661- 3.204
024-XXX	---	---	4.3 -10.5	3.540	2.680-2.227	1.602- 2.657	2.661- 3.204
025-XXX	---	---	7.0 -11.0	5.790	8.462-5.952	.489- .739	.632- .900
026-XXX	---	---	7.0 -11.0	5.790	7.802-5.487	.489- .739	.632- .900
027-XXX	---	---	7.0 -18.0	6.120	8.661-5.121	.501- .951	.618- 1.017
028-XXX	---	---	7.0 -18.0	6.120	8.772-5.186	.501- .951	.611- 1.003
029-XXX	---	---	7.0 -18.0	6.120	8.087-4.782	.501- .951	.611- 1.003
030-XXX	---	---	7.05-10.0	5.790	7.055-6.413	.496- .868	.792- .872
031-XXX	---	---	7.05-10.0	5.790	6.505-5.913	.496- .868	.792- .872
032-XXX	---	---	7.5 -18.0	6.940	12.89-5.589	.335- .815	.394- 1.294
033-XXX	---	---	7.5 -18.0	6.940	11.88-5.153	.335- .815	.394- 1.294
034-XXX	---	---	8.2 -12.4	6.561	10.47 -9.254	.366- .511	.421- .479
035-XXX	---	---	8.2 -12.4	6.561	9.653-8.532	.366- .511	.421- .479
036-XXX	---	---	8.2 -12.4	6.561	9.653-8.532	.366- .511	.420- .465
037-XXX	---	---	8.2 -12.4	6.561	9.653-8.532	.366- .511	.420- .465

1/ At 20°C waveguide temperature.

2/ These values were determined by calculating the unpressurized air dielectric breakdown strength produced by considering the E field within the waveguide. This determination was based on the peak value of a continuous wave (cw) signal. For further information or to calculate for other situations, see Gould and Gilden's "Handbook of High Power Capabilities of Waveguide Systems" (available from Microwave Associates, Burlington, Massachusetts).

3/ These values were determined by calculation of the rate of heat loss to the ambient, considering a nonpressurized air dielectric waveguide in air using no artificial heat sink. A maximum waveguide temperature of 71°C, 1:1 VSWR, and an ambient temperature of 30°C were assumed. For further information or to calculate for other situations, see H.E. King, "Rectangular Waveguide Theoretical CW Average Power Rating", IRE Transactions PGMTT-9 pp 349-357, July 1961.

4/ These waveguides will propagate the TE_{20} mode at frequencies above 4.160 GHz.

5/ These waveguides will propagate the TE_{20} mode at frequencies above 8.570 GHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:44Z
Check the source to verify that this is the current version before use.

TABLE III. Typical resistivity and conductivity of materials.

Material	Resistivity @ 20°C (max)	Conductivity
	microhm-cm	percent
Aluminum:		
Alloy 6061	4.00	40 to 45
Alloy 6063	3.40	50
Copper:		
OF	1.77	101
DLP	1.793	up to 100
Copper alloy	3.9	44

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0952)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-X-5174C

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1426 1 OCT 76 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:44Z
Check the source to verify that this is the current version before use.